However, gravity can surmount that one-way barrier, Cooper discovers. Gravitational signals can go backward in time from Cooper to Murph. We first see this when Cooper desperately pushes books out of Murph's bookcase. Figure 30.1 shows a still from that scene of the movie.



Fig. 30.1. Cooper pushes on the world tube of a book with his right hand. *[From* Interstellar, *used courtesy of Warner Bros. Entertainment Inc.]*

To explain this still, I must tell you a bit more about the bedroom extrusions, as Chris and Paul Franklin explained them to me. Let's focus on the front blue extrusion in Figures 29.10 and 29.12, which I reproduce as Figure 30.2 with extraneous stuff removed. Recall that this extrusion is a set of vertical cross sections through Murph's bedroom, traveling forward in bedroom time along the blue direction (rightward).